

Assignment 4
Assigned on 3/31/2026
Due on Canvas, on 4/9/2026 11:59PM
(50 points)

Fault Diagnosing Smart Home System using ProbLog

Write a simple program using **ProbLog**. The goal of this assignment is to model and reason about uncertainties in a **smart home fault diagnosis system**. The idea is to

- (1) build a knowledge base that represents the probabilistic dependencies among various components of a smart home and
- (2) use it to diagnose likely faults when certain symptoms are observed.

We seek to:

1. Understand how to model uncertainty using probabilistic logic programming.
2. Gain experience with **ProbLog** syntax and semantics.
3. Learn to reason with uncertain observations and causes.
4. Practice debugging and interpreting probabilistic queries

Smart homes have many interconnected components (thermostats, lighting systems, motion sensors, door sensors, etc.).

Each of these components can be in one of the two states - fail, not fail. Therefore, each can be represented as a Boolean Variable.

Failures of some components may affect other components.

However, due to sensor noise or incomplete observations, the system must reason probabilistically about the actual state of the world.

ProbLog allows encoding such uncertain knowledge using **probabilistic facts** and **logical rules**.

Consider a simplified smart home system with the following components:

- **Motion sensors** in the living room and hallway: they report if motion detected or not
- **Light sensors**: they report if the light is on or off.
- **Temperature sensor**: Occasionally it may fail.
- **Thermostat**: can fail to turn on the heating system.
- **Power supply**: can occasionally be unstable.

You will encode a ProbLog model representing the relationships and failure probabilities between these components.

TASKS

Task 1. Model the Smart Home System in ProbLog:

- Define probabilistic facts for component failures. For example:

```
0.05::faulty_motion_sensor(living_room).
0.05::faulty_motion_sensor(hallway).
0.02::faulty_light_sensor.
0.03::faulty_temp_sensor.
0.01::faulty_thermostat.
0.01::power_failure.
```

- Define rules for deriving observations, such as:

```
motion_detected(Room) :-
    not(faulty_motion_sensor(Room)),
    movement(Room).
```

```
light_on :-
    not(faulty_light_sensor),
    light_switch_on.
```

```
heating_on :-
    not(faulty_thermostat),
    not(power_failure).
```

- Encode possible causes for lack of heating, incorrect temperature readings, lights not turning on, etc.

Task 2: Define Queries

- Given observations, query the probabilities of faults:

```
evidence(motion_detected(living_room), false).
evidence(light_on, false).
query(faulty_motion_sensor(living_room)).
query(power_failure).
query(faulty_light_sensor).
```

Task 3: Write a Report

- Explain how your ProbLog program models the system.

- Interpret the results of several queries with **different sets of evidence**.
- Discuss how changing the prior probabilities affects inference.

DELIVERABLES

1. A ProbLog file (**GroupNameSmart_home.pl**) containing:
 - Probabilistic facts
 - Logical rules
 - At least 5 interesting queries
2. A short report (**GroupLeaderNamereport.pdf** or .docx) with:
 - System modeling explanation
 - Sample queries and outputs
 - Interpretation and discussion